1. Clinical Question & Core Hypothesis

Clinical Question

PSA-based metrics (including PSA density) can miss or misclassify aggressive prostate tumors—especially when PSA is pharmacologically suppressed—raising the question: what tumor-intrinsic biology decouples PSA signals from true disease burden and alters susceptibility to immune- or PSMA-directed therapies? 2) Tier 1 paths used: (a) PSA density/5α-reductase inhibitor path highlights how PSA signals can be masked and motivates seeking orthogonal biomarkers; (b) PSMA expression in prostate cancer stem cells provides a molecular axis that could sustain occult, therapy-resistant tumor reservoirs; (c) TMPRSS2/ERG → PI3K → PD-L1 links oncogenic signaling to immune evasion, mechanistically connecting tumor biology to the immune microenvironment. I built the hypothesis by linking the PSMA–PD-L1 immune-evasive axis to disrupted lymphoid organization. 3) Tier 2 support: PSA density's role as a reclassification predictor and PSA as a dominant clinical factor justify focusing on cases where PSA-driven decisions fail; BCG/urothelial immune-engagement provides an external example that local immune structures (TLS/TIL) matter for therapy response. 4) Endpoint choice (therapy_response): PD-L1 and PSMA are actionable targets (immune-checkpoint and PSMA-directed therapies); asking whether this axis predicts or explains poor response is translationally urgent and testable. 5) Deliberate exclusions: To keep the hypothesis sharp I excluded TMPRSS2/ERG and PI3-kinase from the core anchor pair (they are placed in secondary_modifiers) even though they form the upstream signaling route; PSA density and 5α-reductase inhibitors are also secondary modifiers rather than core anchors to avoid exceeding the 2-anchor limit. I chose TLS-like lymphoid aggregates (rather than generic TIL density) because the mechanistic question concerns organized local immune structures that PD-L1–driven suppression could disrupt?

Working Hypothesis

Mechanistic claim: In PSMA-expressing prostate cancer, PD-L1–mediated immune suppression disrupts the formation or maintenance of TLS-like lymphoid aggregates in the tumor microenvironment, decoupling PSA-density signals from true tumor immunogenicity. Predicted clinical consequence: Tumors with high PSMA and PD-L1 expression but low or discordant PSA density will be less likely to respond to PD-1/PD-L1–targeted immunotherapies and PSMA-directed therapies (therapy_response) due to impaired TLS-driven immune engagement.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PSMA, PD-L1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PSMA: PSMA-directed therapies? 2) Tier 1 paths used: (a) PSA density/5α-reductase inhibitor path highlights how PSA signals can be masked and motivates seeking orthogonal biomarkers; (b) PSMA expression in prostate cancer stem cells provides a molecular axis that could sustain occult, therapy-resistant tumor reservoirs; (c) TMPRSS2/ERG → PI3K → PD-L1 links oncogenic signaling to immune evasion, mechanistically connecting tumor biology to the immune microenvironment.

PD-L1: PD-L1 and PSMA are actionable targets (immune-checkpoint and PSMA-directed therapies); asking whether this axis predicts or explains poor response is translationally urgent and testable.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PSA density	Supporting modifier from knowledge graph
Secondary	5α-reductase inhibitors	Supporting modifier from knowledge graph
Secondary	TMPRSS2/ERG	Supporting modifier from knowledge graph
Secondary	PI3-kinase pathway	Supporting modifier from knowledge graph
Secondary	prostate cancer stem cells	Supporting modifier from knowledge graph
Secondary	PSA levels	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective analyses correlating TLS presence, PSMA/PD-L1 status, and PSA density with therapy_response across PD-1/PD-L1– and PSMA-directed therapies;

Prediction 2: prospective biopsies and imaging to quantify TLS-like structures and biomarker expression;

Prediction 3: in vitro/ex vivo models (organoids or co-cultures with autologous immune cells) to probe the effect of PD-L1 signaling on TLS-like organization;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PSA density: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- 5α-reductase inhibitors: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.